

LESSON PLAN	COURSE: MHF4U
Date: Lesson #6	Unit of Study: Chapter #3 Rational Functions
Materials / Resources: - Advanced Function 12 Study Guide - Chapter #3 Formula, Aid and Homework Sheet(s) - KWL Chart	Lesson Focus - Solve Rational Equations and Inequalities - Making Connections with Rational Functions and Equations
Prior Knowledge: - Rational functions and their characteristics - Able to solve equations and finding roots (x-intercepts)	
Overall Expectations: 1. Identify and describe some key features of the graphs of rational functions, and represent rational functions graphically 2. Solve problems involving polynomial and simple rational equations graphically and algebraically 3. Demonstrate an understanding of solving polynomial and simple rational inequalities	
Specific Expectations: 3.5 Determine, through investigation using technology (e.g., graphing calculator, computer algebra systems), the connection between the real roots of a rational equation and the x-intercepts of the graph of the corresponding rational function, and describe this connection [e.g., the real root of the equation $\frac{x-2}{x-3} = 0$ is 2, which is the x-intercept of the function $f(x) = \frac{x-2}{x-3}$; the equation $f(x) = \frac{1}{x-3}$ has no real roots, and the function $f(x) = \frac{1}{x-3}$ does not intersect the x-axis] 3.6 Solve simple rational equations in one variable algebraically, and verify solutions using technology (e.g., using computer algebra systems to determine the roots; using graphing technology to determine the x-intercepts of the graph of the corresponding rational function) 3.7 Solve problems involving applications of polynomial and simple rational functions and equations [e.g., problems involving the factor theorem or remainder theorem, such as determining the values of k for which the function $f(x) = x^3 + 6x^2 + kx - 4$ gives the same remainder when divided by $x - 1$ and $x + 2$] 4.1 Explain, for polynomial and simple rational functions, the difference between the solution to an equation in one variable and the solution to an inequality in one variable, and demonstrate that given solution satisfy an inequality (e.g., demonstrate numerically and graphically that the solution to $\frac{1}{x+1} < 5$ is $x < -1$ or $x > -\frac{4}{5}$) 4.2 Determine solutions to polynomial inequalities in one variable [e.g., solve $f(x) \geq 0$, where $f(x) = x^3 - x^2 + 3x - 9$] and to simple rational inequalities in one variable by graphing the	

corresponding functions, using graphing technology, and identifying intervals for which x satisfies the inequalities

Learning Goals:

- Students will be able to solve for rational equations using the properties of polynomial equations
- Students will be able to solve for simple rational inequalities using the properties of polynomial equations
- Students will be able to determine and graph special cases of simple rational functions through investigation
- Students will be able to solve for contextual problems involving simple rational functions and equations

Success Criteria:

1. By the end of this lesson I will be able to solve rational equations using the properties of polynomial equations
2. By the end of this lesson I will be able to solve simple rational inequalities using the properties of polynomial equations
3. By the end of this lesson I will be able to determine and graph special cases of simple rational functions through investigation
4. By the end of this lesson I will be able to solve contextual problems involving simple rational functions and equations

Assessment Strategies:

Formative: homework for chapters 3.4 & 3.5

Summative: test #2

Time:**Part 1****Introduction:**

Take up / check homework from lesson #5 and check for student's understanding. If needed, go over specific questions.

Time:**Part 2****Lesson:**Chapter 3.4 Solve Equations and Inequalities

Going through the characteristics of a simple rational equation, determine the key points. Start by factoring the expression in the numerator and denominator to find asymptotes and restrictions. Ensure that one side is always equal to zero. Acknowledge that the factored numerator determines the x -intercepts, as this is the only way for $y = 0$ and the factored denominator determines the vertical asymptotes as they are the restricted x -values since the denominator cannot equal to zero. Create an interval with all the significant values and through inputting numbers within each interval, determine the sign for the whole function, whether it is positive or negative.

Chapter 3.5 Making Connections with Rational Functions and Equations

	Recognize the definition of a discontinuity when there is only one single point. Explain that this occurrence only happens when a factored term within the denominator is canceled with a factored numerator. Since that the graph equals to the reduced version, we still need to acknowledge the discontinuity that the canceled factor term made since that was the <i>original</i> function. (e.g. $\frac{2-x}{(2-x)x}$ reduces to $\frac{1}{x}$, however there is a hole at $x = 2$ for the function $\frac{1}{x}$). Introduce the new kind of asymptote which follows an equation that's similar to $y = x$. Identify the procedure of finding this type of asymptotes, which follows by a long division where the numerator is divided by the denominator. The remainder of this procedure is the oblique asymptote. Define the shape of the function by then again plugging in numbers within each interval that is bounded by significant values of the function. Note that the function is not always "within" 2 asymptotes, as they could sometimes cross the asymptotes.
Time: Part 3	Wrap-up: Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions. Use KWL chart to better understand student's learning needs for chapter #3.
Notes, reminders & homework: <u>Reminder</u> - Quiz #3 (to be written before / after current class) - Test #2 (to be written before / after lesson #8) <u>Homework:</u> Chapter 3.4: #1, 2, 4 – 11, 13, 15 Chapter 3.5: #2, 4 – 8, 12	
Lesson Reflection:	

